Risk Perspectives in Distributed AI Systems for Complex Environments

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Abstract

This paper introduces an operational perspective on risk in multi-agent AI systems in complex environments. The analysis focuses on issues and constraints that typically arise in real-world applications. It considers dependencies and influences between agents and how slow and fast processes, transient and steady state physical phenomena might have impact. The paper discusses the necessity of context-based decisions in complex and critical environments. The estimation of the current context of process, system, and environment are all influencing the overall risk assessment and safety. Internal and external risk perspectives are introduced. In addition, a workflow perspective covering design, testing, and operational stages is considered. An approach is introduced to enable risk mitigation.

1 Introduction

With the current trend of increasingly multi-agent AI systems, it becomes attractive to consider how such systems can be used for large-scale automated decision-making. The motivation is usually to increase efficiency and safety in certain domains and processes. Examples include a wide range of applications from the public sector [Stapleton *et al.*, 2022; Black *et al.*, 2022] to justice systems [Zilka *et al.*, 2022], the transport domain [Benedikt *et al.*, 2024] and industrial automation in high-risk processes with dynamic environments [Mihai *et al.*, 2022]. Anticipating harms and responding to them when such systems are in use was investigated in [Chan *et al.*, 2023].

Causal understanding and dependencies between agents, the nature of the process, and the environment in which the system operates are all aspects that influence the risks associated with such systems. When dealing with complex environments, one might be faced with observability issues, delays in measurements from sensors, uncertainties due to sensor quality or measurement process. At the same time, the contextual perspective in fast developing high-risk processes, if ignored, may lead to potentially damaging situations. A very common situation, when failures occur when such automated decision-making AI systems are used in real-world settings, is due to the fact that the complexity of the real industrial systems was underestimated during the development [Gurdur Broo, 2026]. Major bottlenecks include the estimation of the current context, the environment and the self-assessment of the current state of system that steers a process. All of these aspects influence the associated risk and are considered in the approach, as briefly introduced in this paper.

An approach, where a distributed observation method was used to map the environment in a holistic fashion, by combining different views of agents, was introduced in [Dansereau *et al.*,]. Modelling the world from an agent limited perspective and using Kalman filters to learn a policy was considered in [Chang *et al.*, 2003]. However, many approaches still assume full observability. In the real-world, this may not necessarily be the case and this is one of the aspects addressed in this paper.

A detailed taxonomy of systemic risks was introduced in [Uuk *et al.*, 2024]. This considers systemic risk elements associated with control, democracy, discrimination, economy, environmental climate change and pollution, fundamental rights, information, security, and irreversible change. Human-AI team trust was investigated in a multidisciplinary perspective with the aim of better understanding trust in human-AI team interactions [Tielman *et al.*, 2025]. Cooperative strategies have been studied in [Conitzer and Oesterheld, 2023; Dafoe *et al.*, 2020], a protocol for agent collaboration in causal learning was introduced in [Mariani *et al.*, 2023] and [Gregorini and Meyer-Vitali,], cooperation with a focus on social dilemmas was approached in [Orzan *et al.*, 2025]. The analysis of failure modes in governed multi-agent systems was addressed in [Reid *et al.*, 2025]. In the case of multi-agent decision making, when limited sensing features are available, Decentralised Partially Observable Markov Decision Processes have been shown to provide promising modelling tools. However, computing costs are high, and this was shown to be NEXP-complete even for the case of 2-agent scenarios. Variations in which the state is jointly fully observable have been investigated in [Melo *et al.*, 2012].

In critical processes, safety is a major concern. Causal models of agents have been used to investigate safety aspects in multi-agent systems. A formal description of agents, which are adapting to changes triggered in the environment by their actions, was introduced in [Kenton *et al.*, 2022]. Further, integration of causal learning and understanding of agent-environment interactions needs to be addressed. A detailed analysis of risks at system level with associated scenarios and a taxonomy of systemic risks were introduced in [Darius *et al.*, 2025]. In critical systems, it is interesting to analyse both the environment and the process-related aspects from an operational perspective. This perspective, together with a holistic view, which integrates system, process and environment level risks for critical processes, are the novel contributions which are briefly investigated in this position paper.

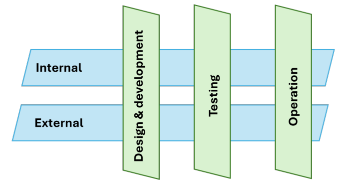

Figure 1: Risk perspectives considered.

2 Risk Perspectives

Here we consider two approaches to risk analysis: the internal and the external risk perspectives. Let's consider a process P and a system S which is designed to steer the process P. The process P is occurring in a physical environment Env. We consider that the process P can be steered through a set of machineries μ, where each machinery involved can automatically execute commands from an external source. Such commands includes updates of operational setpoints. The system S decides what actions to take to control the process P, according to its defined objective. At the same time, the system S provides corresponding setpoints to steer the machinery μ available. In this setting, the internal risk perspective is defined as the set of risk elements associated with the system S involved. The external risk perspective is defined as the set of risk elements which are associated with the environment Env and the process P. The internal and external risk elements are not necessarily independent [Mihai *et al.*, 2022; Gurdur Broo, 2026]. At the same time, an evaluation of the risk elements might not be static and could evolve with local contextual conditions and over time. All of these aspects are aimed at being included in the approach presented in this paper. The approach represents an operational perspective, focusing on issues and constraints that usually arise in real-world applications.

The risk perspectives considered here address elements corresponding to the system S, the environment Env, and process P directions. The aspects considered include environment complexity, problem complexity and the estimation of the current state of the environment. The latter is particularly relevant and potentially difficult to achieve since critical environments are defined by uncertainty of conditions, dynamics, and observability issues [Gurdur Broo, 2026; Cayeux, 2019; Cayeux *et al.*,]. To cope with some of these challenges, some of the models used to characterise the physical phenomena might not capture crucial details, which will further influence the overall risk picture.

Internal and external elements in monolithic versus distributed/multi-agent architectures will impact the overall risks. For instance, training of the underlying algorithms will heavily depend on the sensor data. Calibration of the sensors, delays in communication of the sensor measurement, uncertainty of the measurements and synchronisation of the measurements coming from different sensors, can impact the quality of the measurement data. This translates automatically into the quality of the models which are trained on these data. How models are capturing the physical phenomena, what approximations are made, running times and storage constraints are also playing a role in the overall risk assessment. Even if the calibration of the sensors is ensured, further calibration may be needed for the sensor data and models. Measurements and how the measurements were performed may influence the data quality. How the data was communicated from sensors may impact the quality as well. For instance, sometimes averaging is done due to bandwidth constraints in communication. Such averaging may make the observation of very fast evolving phenomena difficult, such as vibrations. Synchronisation of the different measurements will also influence how the data can be used in further estimations of the current context. How causal understanding is included in the architecture, will impact the estimations and operational use.

In addition, to the internal and external perspectives, we consider a workflow perspective which comes across. The workflow perspective addresses risks associated with the different stages, namely design, testing and operational, Figure 1. Each of these can have elements from the internal and external perspectives mentioned above.

In this paper, we do not consider that an agent might be selfish and avoid sharing information on purpose. We therefore assume that the AI agents involved are behaving in a truthful manner and acting to the best of their knowledge. Hence, any miscoordination or conflict will arise from other causes, such as different estimations that the agents are performing on the environment, process, or their own and each other's state. In addition, limitations on communication due to constraints, such as space and time, are outside of the focus of this paper. For more details and an analysis of the risks associated with communication constraints, see [Hammond *et al.*, 2025].

Adaptive agents usually assume a common understanding of the environment. In real-world applications, estimating the status of the environment and achieving a model for the environment might not be a trivial task and could easily lead to different assumptions and, hence, different models. We remove this assumption here and allow for different views for the different AI agents. In addition, it is relevant to consider that an AI agent might be using different models, based on the context of the process. For instance, in an automated system for a critical operation, an AI agent might be using a model focusing on vibrations estimation at a certain stage of the operation, while at other stages, models focusing on heat transfer estimation may be more relevant. While considering combining the two aforementioned models into one large model may be attractive, it is not always straightforward in practice. This can be due to computational constraints or a trade-off on accuracy.

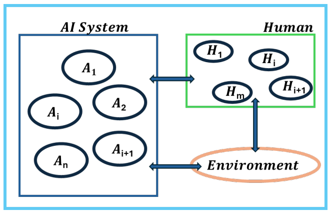

Figure 2: A schematic of a multi-agent AI system, human agents and physical environment.

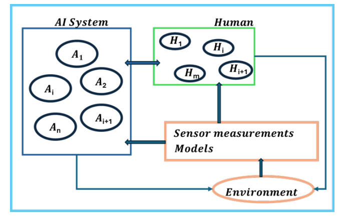

Figure 3: A schematic of a multi-agent system, human agents and physical environment in a monolithic perspective.

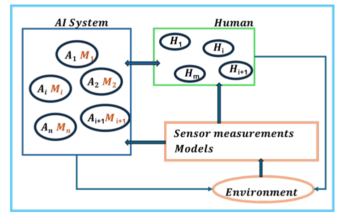

Figure 4: A schematic of a multi-agent system, human agents and physical environment in a distributed perspective.

3 A Generic Approach to Enable Operational Risk Mitigation from Architecture Design

We propose an approach in which the environment is tightly included in the overall architecture with the different perspectives and estimations corresponding to the AI agents involved. By environment we mean the physical environment. By considering the dependencies between environment perspectives and the rest of the system, AI agents and human agents can contribute to the understanding and the online evaluation of the causal relations in the overall distributed system. Even though the agents share the same physical environment, the impact of the actions of different agents on the environment might be different. At the same time, in complex environments, some actions might have a medium- or long-term effect which may be difficult to capture in the models aiming to estimate the environment evolution. By allowing and including different environment perspectives, one can allow to define and include in the control process possible effects of such dynamics. In real-world applications, the view of the environment might be different for the different AI systems that make up the overall system in Figure 2.

In the approach considered here, we assume that all the sensor measurements are made available simultaneously to all AI agents involved in the process.

We are discussing two different perspectives: a monolithic perspective and a distributed perspective. In the monolithic perspective, we assume that all AI systems and agents have access to the same description of the physical environment. Even in this situation discrepancies might arise due to timing differences or variations in sensors and measurements. Any such discrepancy can lead to potentially safety related issues given the context considered here where the AI agents can take and execute decisions in a critical environment.

We consider further a scenario where several sensors are available to measure certain physical processes in the environment. We further assume that all the measurements are made available to all AI systems and agents at the same time on a common layer, see Figure 3.

In real-world processes, as discussed in previous sections, not all physical phenomena can be measured, thus it might be necessary to use either methods such as sensor fusion or other models to estimate these. If several solutions are available for these, we assume that these are also available through the common layer, as illustrated in Figure 3.

In the distributed perspective, we assume that the AI systems and agents involved have their own estimations of the environment. Although sensor data are assumed to be commonly shared between all AI systems and agents, specific models might not be commonly shared with the rest of the system; see Figure 4. This adds a new level of complexity to the overall approach. When such internal specific models are used by the systems to compute actions or provide data to the rest of the system, it can lead to potential inconsistencies. We propose to enable sharing between agents, the information related to human interaction, and environmental perspectives.

Continuous sharing of information on elements which are focused on in the internal modelling of an agent can provide the rest of the AI system with the capability to detect possible inconsistencies and potentially trigger mitigating actions. Common calibration of the models based on data and communication with other agents can also contribute to an operational alignment of the internal views of the agents. Centralised calibration of the data measurements and other data sources is needed to ensure a consistent starting point towards estimation of current state of the environment. Consider a scenario in which an agent A_i is computing the current constraints associated with the process. In addition, consider that two other agents A_j, M_j and A_k, M_k are using these constraints to compute current operational setpoints. Different internal models M_j and M_k will translate into different operational setpoints which may lead to dangerous situations in real-world applications. By sharing information on the limits of the associated models, it could enable coordination between the different agents and potentially lead to increase horizon knowledge for the individual agents. In addition, mutual trust between agents may be enabled when enough information is shared and potentially validated during operation.

4 Discussion

In our analysis, we assumed that the data measurements are commonly shared with all the AI agents and systems. If this is not the case and different agents might have access to different data, coordination methods can be used to centralise all information and enable common sharing to all agents. However, it is interesting to explore how the online coordination process might influence the overall process and corresponding risk.

In the approach described here, we assume a trust-based relation between the AI agents [Sapienza *et al.*, 2026], meaning that a priori all AI agents aim to contribute to the overall goal or individual goals to the best of their knowledge. However, this approach does not exclude potential conflicts that might arise when individual goals might be different than the overall goals for instance, or when internal modelling capabilities that an agent is relying on are not appropriate to estimate a certain process or environment state.

Yet another further direction is to consider how to cope with situations where agents may intentionally try to disturb the process by providing erroneous information. Consequently, it is interesting to assess what protection mechanisms should be in place to address these situations. Possible solutions may include dynamic verification and validation mechanisms and safety features, such as online estimations of limits to ensure safe operations. It is also interesting to further explore how mutual trust between AI agents may be achieved, maintained, and assessed throughout the process.

5 Conclusion

This paper introduces an operational perspective on risk in multi-agent AI systems in complex environments, where internal and external factors are considered. The approach here focuses on issues and constraints arising in real-world applications and in critical domains. The paper aims to underline several crucial aspects to be considered when assessing factors influencing the risk and safety related aspects when AI systems are controlling processes in critical environments. Examples include the estimation of the current context of process, system and environment which are all influencing the overall risk picture. In addition to the internal and external perspectives, a workflow perspective covering design, testing and operational stages are briefly introduced. The aim is that such an approach can pave the way towards self-calibrating causal models which account for current context of the system, process and environment and thus contributing to increase safety when in use.

Ethical Statement

There are no ethical issues.

Acknowledgments

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